

A Literature Survey on Intelligent Reflecting Surface based Wireless Communications

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May 13, 2020

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Introduction

**Towards Smart and
Reconfigurable Environment:
Intelligent Reflecting Surface
Aided Wireless Network -
Qingqing Wu and Rui Zhang**

Motivation

- Ubiquitous wireless connectivity for at least 100-billion devices and 1000-fold network capacity increase largely achieved by 5G wireless networks
 - mmWave and massive MIMO
 - Ultra dense network (UDN)
- But several bottlenecks
 - High complexity and increased hardware cost
 - Energy consumption issues
 - Network interference issues in UDN
- Research on finding spectral and energy efficient solutions for beyond 5G is imperative
- Intelligent reflecting surface (IRS), a promising new technology
 - Reconfigures the wireless propagation environment via software controlled reflection

Intelligent Reflecting Surface

- IRS is a planar surface comprising a large number of passive reflecting elements
 - Each element induces an amplitude/phase change to the incident signal
 - Collaboratively achieves 3D reflect beamforming
- Proactively modifies the wireless channel between the Tx and Rx via signal reflections
 - Additional degree of freedom to improve the wireless communication performance
- Can be fabricated to mount on arbitrarily shaped surfaces to cater to different application scenarios

Typical IRS Applications in Wireless Network

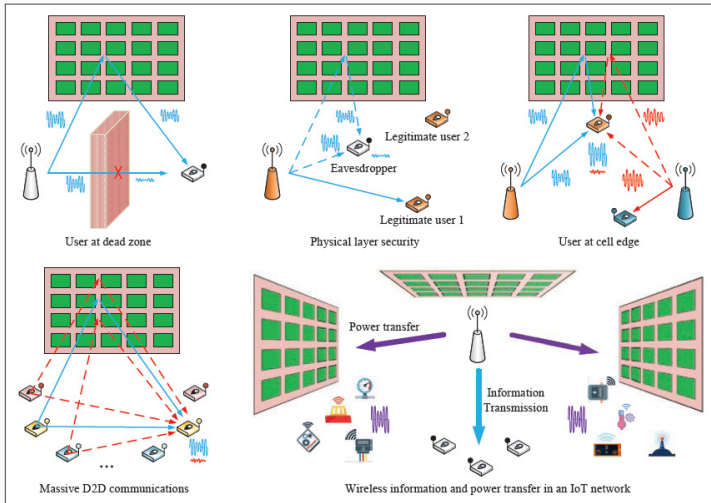


Figure 1. Typical IRS applications in wireless network.

Signal Model

- Three components: BS-IRS link, IRS's reflection and IRS-UE link
- Superposed signal from the transmitter reflected by each element of IRS with different adjustable amplitude and phase shifts
 - Multiplicative channel model
- Reflected signal by the n^{th} element of IRS is given by

$$y_n = \beta_n \exp(j\theta_n)x,$$

where $y_n \in [0, 1]$ and θ_n specify the reflection coefficient, x is the incident signal

- Adjust the reflection coefficients to control the reflected signal
- Discrete amplitude and phase shift model for cost effectiveness

Hardware Architecture

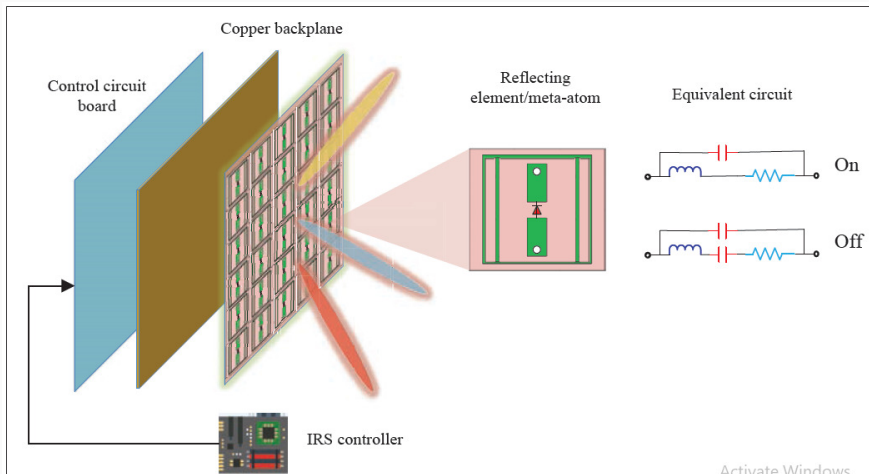


Figure 2. Architecture of IRS.

Activate Windows
Go to Settings to activate Wi

Main Design Challenges

- Passive Beamforming design
- Channel acquisition
- Deployment

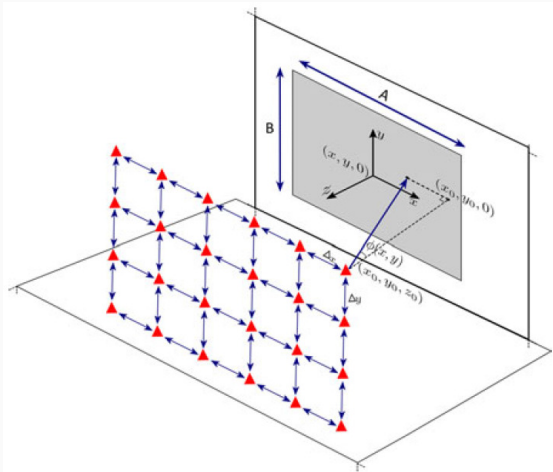
Literature Survey

**Beyond Massive MIMO: The
Potential of Data Transmission
with Large Intelligent Surfaces -
Sha Hu, Fredrik Rusek, and Ove
Edfors**

Contributions

- Studied the information-transfer capabilities of the LIS in the uplink
- Considered only line of sight propagation for analytical tractability
- Unlike massive MIMO systems, the information theoretic analysis considers near field effects (but most papers covered ignore this effect except this one)
- Analyzed the normalized capacity ($\hat{C}$) per volume unit in space
- Analyzed the spatial degrees of freedom (DoF) that can be harvested with a LIS (prelog factor) when the deployed surface area is infinitely large
 - $2/\lambda$ terminals can be spatially multiplexed per meter deployed surface for a 1-D terminal deployment
 - π/λ^2 terminals can be spatially multiplexed per m^2 deployed surface area for two and three dimensional terminal deployments

System Model



- Transmission from K user terminals in a 3-D space to a 2-D LIS
- Consider LoS propagation for analytical tractability (more reasons given in the paper)

Narrow-Band Received Signal Model at the LIS

- Received baseband signal at the LIS location $(x, y, 0)$ corresponding to the k^{th} terminal located at (x_k, y_k, z_k) at time t is given as

$$\begin{aligned} \tilde{s}_{x_k, y_k, z_k}(x, y, t) = & s_{x_k, y_k, z_k}(x, y) \sqrt{P_k} \sum_{m=-\infty}^{\infty} u_k[m] \\ & \times \text{sinc}_T(t - mT) + n(x, y, t), \end{aligned} \quad (1)$$

- The effective channel can be modeled as

$$s_{x_k, y_k, z_k}(x, y) = \sqrt{\varepsilon_L \cos \phi(x, y)} \exp(-2\pi j f_c \Delta_k(x, y)), \text{ where}$$

- η_k is the square of the distance from the k^{th} UE to the location $(x, y, 0)$ at the LIS: $\eta_k = z_k^2 + (y - y_k)^2 + (x - x_k)^2$
- $\varepsilon_L = 1/(4\pi\eta_k)$ denotes the free space attenuation
- $\phi(x, y)$ is the AoA with $\cos \phi(x, y) = z_k / \sqrt{\eta_k}$
- $\Delta_k(x, y) = \sqrt{\eta_k}/c$ is the transmit time
- Effective channel after inserting the parameters:

$$s_{x_k, y_k, z_k}(x, y) = \frac{\sqrt{z_k}}{2\sqrt{\pi}\eta_k^{3/4}} \exp\left(-\frac{2\pi j \sqrt{\eta_k}}{\lambda}\right). \quad (2)$$

Rx Signal Model for Multiple Terminals with MF Procedure

- Rx signal at the LIS location $(x, y, 0)$ of signals from all K UEs

$$r(x, y, t) = \sum_{k=0}^{K-1} \sum_{m=-\infty}^{\infty} s_{x_k, y_k, z_k}(x, y) \sqrt{P_k} u_k[m] \text{sinc}_T(t - mT) + n(x, y, t).$$

- After MF, the rx signal at sampling time mT is

$$r_k[m] = \sum_{\ell=0}^{K-1} \sqrt{P_k P_\ell} \phi_{k, \ell} u_\ell[m] + w_k[m], \quad (3)$$

where $\phi_{k, \ell} = \iint_{(x, y) \in \mathcal{S}} s_{x_\ell, y_\ell, z_\ell}(x, y) s_{x_k, y_k, z_k}^*(x, y) dx dy$, and $\mathcal{S}$ is the surface-area spanned by the 2-D LIS

- In matrix form, $\mathbf{r} = \mathbf{G}\mathbf{u} + \mathbf{w}$, where the $(\ell, k)^{\text{th}}$ element of $\mathbf{G}$ equals $g_{k, \ell} = \sqrt{P_\ell P_k} \phi_{k, \ell}$, and $\mathbb{E}[\mathbf{w}\mathbf{w}^H] = N_0 \mathbf{G}$.
- Equal power transmission assumed in the rest of the paper

Independent Signal Dimensions Harvested with the LIS

- Channel capacity averaged by the number of terminals (nats/s/Hz):
 $\mathcal{C} = \frac{1}{K} \log \det(\mathbf{I} + \frac{\mathbf{G}}{N_0}).$
- Consider the LIS as a large 2-D antenna array
- For 1-D and 2-D terminal deployments, the space normalized capacity (nats/s/Hz/m) are

$$\hat{\mathcal{C}} = \lim_{K \rightarrow \infty} \frac{K\mathcal{C}}{K\Delta_x} = \lim_{K \rightarrow \infty} \frac{\mathcal{C}}{\Delta_x}, \quad (4)$$

$$\hat{\mathcal{C}} = \lim_{K \rightarrow \infty} \frac{K\mathcal{C}}{K\Delta_x\Delta_y} = \lim_{K \rightarrow \infty} \frac{\mathcal{C}}{\Delta_s} \quad (5)$$

- The number of independent signal dimensions

$$\rho = \lim_{\hat{P}/N_0 \rightarrow \infty} \frac{\hat{\mathcal{C}}}{\log(\hat{P}/N_0)}, \quad (6)$$

where $\hat{P} = P/\lambda$ and $\lambda = \Delta_x$ or Δ_s

Space-Normalized Capacities and Independent Signal Dimensions

- Property used to derive the capacity results for 1-D UE deployment:
For sufficiently small λ , the integral

$$g(\Delta) = \int_{-\infty}^{\infty} (1+x^2)^{-\frac{3}{4}} (1+(x+\Delta)^2)^{-\frac{3}{4}} \\ \times \exp\left(-\frac{2\pi j}{\lambda} \left[\sqrt{1+x^2} - \sqrt{1+(x+\Delta)^2}\right]\right) dx, \quad (7)$$

can be well approximated by a sinc function

$$g(\Delta) \approx 2\text{sinc}\left(\frac{2\Delta}{\lambda}\right). \quad (8)$$

- Capacity results derived for 1-D, 2-D and 3-D terminal deployments

**Weighted Sum-Rate
Maximization for Intelligent
Reflecting Surface Enhanced
Wireless Networks - Huayan
Guo, Ying-Chang Liang, Jie
Chen, Erik G. Larsson**

Contributions

- Investigated the weighted sum-rate (WSR) maximization problem for the MU downlink (DL) MISO system aided by IRS
- Active beamforming (BF) at the BS and passive BF at the IRS are decomposed based on the Lagrange dual transform
- Active BF solved using the multi-ratio quadratic transform and passive BF reformulated as QCQP and solved using ADMM

System Model

- Rx signal at user k is: $y_k = (\mathbf{h}_{d,k}^H + \mathbf{h}_{r,k}^H \mathbf{\Theta}^H \mathbf{G}) \sum_{k=1}^K \mathbf{w}_k s_k + u_k$
- SINR of user k : $\gamma_k = \frac{|\mathbf{h}_{d,k}^H + \mathbf{h}_{r,k}^H \mathbf{\Theta}^H \mathbf{G}|^2 |\mathbf{w}_k|^2}{\sum_{i=1, i \neq k}^K |\mathbf{h}_{d,i}^H + \mathbf{h}_{r,i}^H \mathbf{\Theta}^H \mathbf{G}|^2 |\mathbf{w}_i|^2 + \sigma_0^2}$.

$$(P1) \quad \max_{\mathbf{W}, \mathbf{\Theta}} \quad f_1(\mathbf{W}, \mathbf{\Theta}) = \sum_{k=1}^K \omega_k \log(1 + \gamma_k)$$

$$\mathbf{s.t.} \quad \theta_n \in \mathcal{F}_D, \quad \forall n = 1, \dots, N,$$

$$\sum_{k=1}^K \|\mathbf{w}_k\|^2 \leq P_T$$

- Applying Lagrange dual transform, problem reformulated as

$$\begin{aligned}
 (\text{P1}') \quad & \max_{\mathbf{W}, \Theta, \alpha} f_{1a}(\mathbf{W}, \Theta, \alpha) \\
 \text{s.t.} \quad & (3a), (3b),
 \end{aligned}$$

where $\alpha = [\alpha_1, \dots, \alpha_K]^T$ and

$$\begin{aligned}
 f_{1a}(\mathbf{W}, \Theta, \alpha) = & \sum_{k=1}^K \omega_k \log(1 + \alpha_k) - \sum_{k=1}^K \omega_k \alpha_k \\
 & + \sum_{k=1}^K \frac{\omega_k (1 + \alpha_k) \gamma_k}{1 + \gamma_k}.
 \end{aligned}$$

- With fixed $\mathbf{W}$ and Θ , optimal $\alpha_k = \gamma_k$

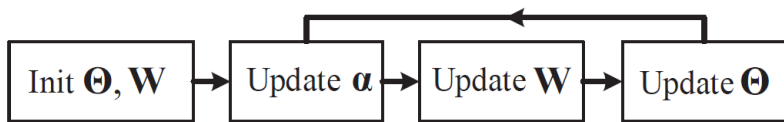


Fig. 2. Alternating optimization for (P1').

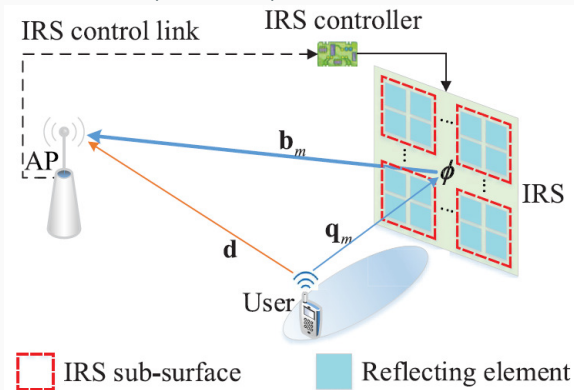
**Intelligent Reflecting
Surface-Enhanced OFDM:
Channel Estimation and
Reflection Optimization -
Beixiong Zheng, and Rui Zhang**

Contributions

- Proposed a transmission protocol to execute channel estimation (CE) and reflection optimization successively for an IRS enhanced OFDM system

System Model

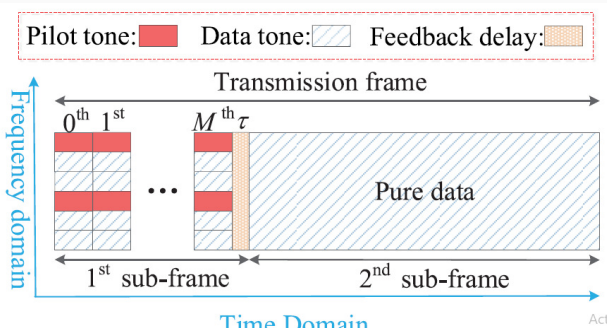
- UL single antenna Tx, Rx system
- During the CE phase (at the AP), full reflection enabled at the IRS



- Received signal after OFDM demodulation:

$$\begin{aligned}
 \mathbf{y} &= \mathbf{X} \left(\sum_{m=1}^M \mathbf{q}_m \phi_m \odot \mathbf{b}_m + \mathbf{d} \right) + \mathbf{v} \\
 &= \mathbf{X} \left(\sum_{m=1}^M \phi_m \mathbf{g}_m + \mathbf{d} \right) + \mathbf{v} \\
 &= \mathbf{X} \underbrace{(\mathbf{G}\phi + \mathbf{d})}_{\mathbf{h}} + \mathbf{v}
 \end{aligned}$$

- Transmission Protocol:



- Pilots transmitted in each OFDM symbol in selected pilot tones using different reflection patterns for each OFDM symbol
- LS channel estimation (of superimposed channel) followed by DFT/IDFT based interpolation

$$\hat{\mathbf{h}}^{(i)} = \mathbf{G}\phi^{(i)} + \mathbf{d} + \bar{\mathbf{v}}^{(i)} = \vec{\mathbf{G}}\vec{\phi}^{(i)} + \bar{\mathbf{v}}^{(i)},$$

where $\vec{\mathbf{G}} = [\mathbf{d}, \mathbf{G}]$ and $\vec{\phi}^{(i)} = \begin{bmatrix} \mathbf{1} \\ \phi^{(i)} \end{bmatrix}$

- By stacking $\hat{\mathbf{h}}^{(i)}$, we get

$$\hat{\mathbf{H}} = \vec{\mathbf{G}}\Theta + \bar{\mathbf{V}}$$

and

$$\begin{bmatrix} \hat{\mathbf{d}} & \hat{\mathbf{G}} \end{bmatrix} = \hat{\mathbf{H}}\Theta^{-1}.$$

- Once channel is estimated, the reflection coefficients can be computed by solving rate maximization problem and fed back to the IRS for data subframe

**Intelligent Reflecting Surface for
Massive Device Connectivity:
Joint Activity Detection and
Channel Estimation - Shuhao Xia
and Yuanming Shi**

Contributions

- Considered an uplink (UL) IRS-assisted IoT network, where a single BS serves massive number of mobile devices
- Formulated the IRS related activity detection and channel estimation as sparse matrix factorization (SMF), matrix completion (MC) and MMV problems
- Proposed a three stage framework based on BiG-AMP for SMF, singular value thresholding (SVT) for MC and VAMP for MMV.

System Model

- Single BS with M antennas, K users, IRS with N reflecting elements. NLoS propagation environment
- Uplink received signal by the BS at the l^{th} symbol duration:

$$\mathbf{y}(l) = \mathbf{G} \left(\mathbf{p}(l) \odot \sum_{k=1}^K \mathbf{h}_k^r \alpha_k \mathbf{x}_k(l) \right) + \mathbf{n}(l)$$

Problem Formulation

$$\mathbf{Y} = \mathbf{G}(\mathbf{P} \odot (\mathbf{H}\mathbf{A}\mathbf{X})) + \mathbf{N}$$

- $\mathbf{G}$ IRS to BS, $\mathbf{A}$ activity matrix (diagonal), $\mathbf{P}$ phase shift matrix, $\mathbf{X}$ known signature matrix
- Define the variables

$$\mathbf{\Theta} = \mathbf{H}\mathbf{A} \in \mathbb{C}^{N \times K}, \quad \mathbf{Q} = \mathbf{\Theta}\mathbf{X} \in \mathbb{C}^{N \times L}, \quad \mathbf{W} = \mathbf{P} \odot \mathbf{Q} \in \mathbb{C}^{N \times L}$$

- Received signal:

$$\mathbf{Y} = \mathbf{G}\mathbf{W} + \mathbf{N}$$

- $\mathbf{P}$ and $\mathbf{W}$ have same sparsity pattern
- Design a sparse $\mathbf{P}$ for recovering $\mathbf{G}$ and $\mathbf{W}$ using SMF

Algorithm Framework

- Recall:

$$\mathbf{\Theta} = \mathbf{H}\mathbf{A} \in \mathbb{C}^{N \times K}, \quad \mathbf{Q} = \mathbf{\Theta}\mathbf{X} \in \mathbb{C}^{N \times L}, \quad \mathbf{W} = \mathbf{P} \odot \mathbf{Q} \in \mathbb{C}^{N \times L}$$

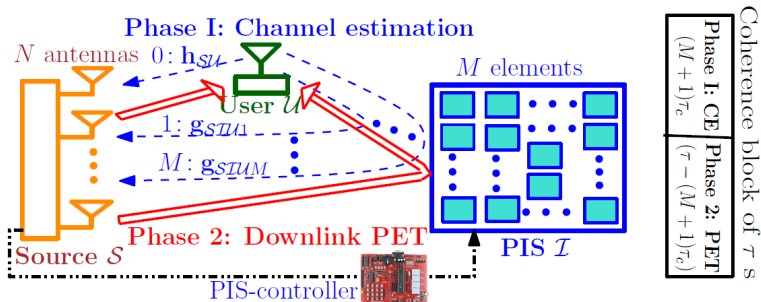
- Step 1: SMF - Recovering the matrices $\mathbf{G}$ and $\mathbf{W}$ from $\mathbf{Y}$. Use Bilinear GAMP (BiG-AMP)
- Step 2: MC: Complete the missing entries of $\mathbf{Q}$ given $\hat{\mathbf{W}}$ and the phase shifting matrix $\mathbf{P}$. Use the low rank structure of $\mathbf{Q}$
- Step 3: MMV recovery: Estimating the matrix $\mathbf{\Theta}$ from $\hat{\mathbf{Q}}$ and known signature matrix $\mathbf{X}$
- Recover the activity matrix $\mathbf{A}$ using thresholding on the ℓ_2 norm of the columns of $\hat{\mathbf{\Theta}}$

**Channel Estimation and
Low-complexity Beamforming
Design for Passive Intelligent
Surface Assisted MISO Wireless
Energy Transfer - Deepak
Mishra; Håkan Johansson**

Contributions

- Proposed a channel estimation (CE) protocol for passive intelligent surface (PIS) assisted energy transfer (PET) from a multiantenna power beacon (PB) to a single antenna energy harvesting (EH) user
- Near optimal closed form solutions for transmit (active) and reflect (passive) energy beamforming (EB) designs derived for PB and PIS for efficient PET

System Model



MISO Wireless Channel Model

- Flat quasi static Rician block fading model assumed
- Channel reciprocity used to estimate the channels and least squares (LS) channel estimation used
- $\mathcal{S}$ -to- $\mathcal{U}$ and $\mathcal{I}$ -to- $\mathcal{U}$ channels are represented as

$$\mathbf{h}_{ik} = \sqrt{\frac{\beta_{ik} K_{ik}}{K_{ik} + 1}} \mathbf{h}_{D_{ik}} + \sqrt{\frac{\beta_{ik}}{K_{ik} + 1}} \mathbf{h}_{S_{ik}},$$

$\forall i \in \{\mathcal{S}, \mathcal{I}\}$, $k \in \mathcal{U}$, and $\mathbf{h}_{D_{SU}}$, $\mathbf{h}_{D_{IU}}$ are deterministic vectors.

$\mathbf{H}_{SI} \in \mathbb{C}^N \times M$ also follows Rician distribution

- Received signal at user is

$$\begin{aligned} y_U &= (\mathbf{h}_{SU}^T + \mathbf{h}_{IU}^T \Theta \mathbf{H}_{SI}^T) \mathbf{f}_A x_e + w_U \\ &= (\mathbf{h}_{SU}^T + \mathbf{f}_P^T \mathbf{G}_{SIU}^T) \mathbf{f}_A x_e + w_U, \end{aligned}$$

where M diagonal entries of the phase shift matrix Θ denote the passive EB vector $\mathbf{f}_p$ at $\mathcal{I}$ and $\mathbf{G}_{SIU} \triangleq \mathbf{H}_{SI} \text{diag}\{\mathbf{h}_{IU}\} \in \mathbb{C}^{N \times M}$

- Uplink pilots of duration $(M + 1)\tau_c$
- ON/OFF phase shifts at PIS: Phase shift matrix (row corresponds to reflecting element index, column corresponds to the time slot)

$$[\Phi]_{i,m} \triangleq \begin{cases} 1 - \epsilon_1, & (i = m) \wedge (m \neq 1) \\ 0 + \epsilon_0, & (i \neq m) \vee (m = 1) \end{cases}, \forall i \in \mathcal{M}, m \in \mathcal{M}_+,$$

where ϵ_1 and ϵ_0 corresponds to the reflector imperfections and $\mathcal{M}_+ \triangleq \mathcal{M} \cup \{M + 1\}$

- Received signal at $\mathcal{S}$ during the CE phase

$$\mathbf{Y}_{\mathcal{S}} = (((\mathbf{h}_{SU} \otimes \mathbf{1}_{1 \times (M+1)}) + \mathbf{G}_{S\mathcal{I}\mathcal{U}}\Phi) \otimes \mathbf{x}_p) + \mathbf{W}_{\mathcal{S}}.$$

- Using the structure of the matrices, $\mathbf{h}_{SU}$ is estimated first followed by $\mathbf{G}_{S\mathcal{I}\mathcal{U}}$
- Joint active and passive energy beamforming coefficients also computed (algorithm in paper) and fed back to the PIS

**An Optimal Channel Estimation
Scheme for Intelligent Reflecting
Surfaces Based on a Minimum
Variance Unbiased Estimator -
Tobias Lindstrøm Jensen,
Elisabeth De Carvalho**

Contributions

- Conventional least squares approach used for channel estimation in a single user SIMO system with passive IRS
- Showed that the estimation performance is dependent on the reflector activation pattern. ON/OFF activation pattern is suboptimal and the rows of the DFT matrix is shown to be the optimal activation pattern

Signal Model

- Single user MISO TDD system. Channel reciprocity used for estimating downlink channels
- Received signal at training step t is

$$\begin{aligned} s_t &= (h_d + G^H \mathbf{diag}(\phi_t) h) x_t + n_t \\ &= (h_d + V \phi_t) x_t + n_t \end{aligned}$$

- In matrix form

$$\Phi = \begin{bmatrix} 1 & \phi_{1,1} & \cdots & \phi_{1,K} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & \phi_{T,1} & \cdots & \phi_{T,K} \end{bmatrix} \in \mathbf{C}^{T \times K+1}, \Psi = \Phi \otimes I_M$$

$$s = \begin{bmatrix} s_1 \\ \vdots \\ s_T \end{bmatrix}, n = \begin{bmatrix} n_1 \\ \vdots \\ n_T \end{bmatrix}, \theta = \begin{bmatrix} h_d \\ v_1 \\ \vdots \\ v_K \end{bmatrix},$$

$$X = \text{diag}([x_1 \mathbf{1}_M, \cdots, x_T \mathbf{1}_M]), H = X\Psi$$

$$\begin{aligned} \begin{bmatrix} s_1 \\ \vdots \\ s_T \end{bmatrix} &= \begin{bmatrix} x_1 [I_M & \phi_{1,1} I_M & \cdots & \phi_{1,K} I_M] \\ & & \ddots & \\ x_T [I_M & \phi_{T,1} I_M & \cdots & \phi_{T,K} I_M] \end{bmatrix} \begin{bmatrix} h_d \\ v_1 \\ \vdots \\ v_K \end{bmatrix} + \begin{bmatrix} n_1 \\ \vdots \\ n_T \end{bmatrix} \\ &= \mathbf{diag}([x_1 \mathbf{1}_M, \cdots, x_T \mathbf{1}_M]) (\Phi \otimes I_M) \theta + n \\ s &= X\Psi\theta + n = H\theta + n \end{aligned}$$

- MVUE is the LS estimator. ON/OFF reflection pattern suboptimal (details in paper)
- Design a reflection pattern by solving an optimization problem

$$\begin{aligned}
& \text{minimize} && \text{tr} (C_{\hat{\theta}}) \\
& \text{subject to} && C_{\hat{\theta}} = \sigma^2 (\Phi^H \Phi)^{-1} \otimes I_M, \Phi \in \mathbf{C}^{T \times K+1} \\
& && [\Phi]_{t,1} = 1, t = 1, \dots, T \\
& && \phi_{t,k} = \beta_{t,k} \exp(j p_{t,k}), t = 1, \dots, T; k = 1, \dots, K \\
& && \beta_{t,k} \in [0, 1], t = 1, \dots, T; k = 1, \dots, K \\
& && p_{t,k} \in \mathbf{L}, t = 1, \dots, T; k = 1, \dots, K.
\end{aligned}$$

- Difficult to solve. Since MVUE attains CRLB, we have the following lower bound per element

$$[C_{\hat{\theta}}]_{i,i} = [\mathcal{I}^{-1}(\theta)]_{i,i} \geq \frac{1}{[\mathcal{I}(\theta)]_{i,i}}$$

- Lower bound attained when the Fisher information matrix is diagonal. Finally results in a unitary matrix as an optimal reflection pattern.

**Intelligent Reflecting Surface
Enhanced Wireless Network via
Joint Active and Passive
Beamforming - Qingqing Wu and
Rui Zhang**

Contributions

- Investigated an IRS enhanced point to point MISO system where the user receives the signals from both the AP and the IRS
- Proposed two algorithms (centralized and distributed) to maximize the total received signal power at the user by optimizing the active and passive beamforming vectors

System Model

- M antenna AP, N element IRS, quasi static flat fading channel, TDD for channel reciprocity
- Rxed signal at the UE: $y = (h_r^H \Theta G + h_d^H) w s + z$
- Optimization problem:

$$\begin{aligned} (P1) : \quad & \max_{w, \theta} \quad |(h_r^H \Theta G + h_d^H) w|^2 \\ & \text{s.t.} \quad \|w\|^2 \leq \bar{p}, \\ & \quad \quad 0 \leq \theta_n \leq 2\pi, \quad \forall n = 1, \dots, N. \end{aligned}$$

Centralized Algorithm

- For a given θ , MRT is the optimal tx beamformer. (P1) reduced to

$$\begin{aligned} \text{(P2): } \quad & \max_{\theta} \quad \|h_r^H \Theta G + h_d^H\|^2 \\ \text{s.t.} \quad & 0 \leq \theta_n \leq 2\pi, \forall n = 1, \dots, N. \end{aligned}$$

- Change of var $h_r^H \Theta G = v^H \Phi$ where $\Phi = \text{diag}(h_r^H)G$, (P2) to (P3)

$$\begin{aligned} \text{(P3): } \quad & \max_v \quad v^H \Phi \Phi^H v + v^H \Phi h_d + h_d^H \Phi^H v \\ \text{s.t.} \quad & |v_n| = 1, \forall n = 1, \dots, N. \end{aligned}$$

- (P3) is a non convex QCQP which can be reformulated as

$$\begin{aligned} \text{(P4): } \quad & \max_{\bar{v}} \quad \bar{v}^H R \bar{v} \\ \text{s.t.} \quad & |\bar{v}_n| = 1, \forall n = 1, \dots, N+1, \end{aligned}$$

$$\text{where } R = \begin{bmatrix} \Phi \Phi^H & \Phi h_d \\ h_d^H \Phi^H & 0 \end{bmatrix}, \bar{v} = \begin{bmatrix} v \\ t \end{bmatrix}$$

- Semidefinite relaxation

$$\begin{aligned}
 (\text{P5}) : \quad & \max_V \quad \text{tr}(RV) \\
 \text{s.t.} \quad & V_{n,n} = 1, \quad \forall n = 1, \dots, N+1, \\
 & V \succeq 0.
 \end{aligned}$$

- Solved for V using SDP and obtained a rank 1 approximation by choosing $\bar{v} = U\Sigma^{\frac{1}{2}}r$, where r is a random vector chosen from $\mathcal{CN}(0, I_{N+1})$. Sufficiently large number of randomizations guarantees a close approximation to the solution of (P3)

Distributed Algorithm

- Distributed implementation of alternating optimization:
 $|(h_r^H \Theta G + h_d^H)w| = |h_r^H \Theta Gw + h_d^H w| \leq |h_r^H \Theta Gw| + |h_d^H w|$ with equality iff $\arg(h_r^H \Theta Gw) = \arg(h_d^H w) = \varphi_0$
- AP sends beamformed pilot to IRS for obtaining $g_n^H w$
- (At the IRS) Apply change of var $h_r^H \Theta Gw = v^H a$ where $v = [e^{j\theta_1}, \dots, e^{j\theta_N}]^H$ and $a = \text{diag}(h_r^H)Gw$, (P1) reduced to

$$\begin{aligned} \text{(P1')} : \quad & \max_v \quad |v^H a| \\ & \text{s.t.} \quad |v_n| = 1, \quad \forall n = 1, \dots, N, \\ & \quad \arg(v^H a) = \varphi_0. \end{aligned}$$

- Optimal solution: $\theta_n^* = \varphi_0 - \arg(h_{n,r}^H) - \arg(g_n^H w)$
- (At the AP) For a given θ , MRT is the optimal transmit beamformer (using the composite channel estimated by uplink pilot from UE)
- Keypoint: To avoid feedback of w to the IRS, w is phase rotated (doesn't change the beamforming gain) such that $\varphi_0 = 0$

Algorithm 1 Distributed Algorithm.

- 1: Initialize ϵ and iteration number $k = 1$.
- 2: Let the IRS be in the receiving mode and the user broadcast a pilot signal. The IRS estimates phase $\arg(\mathbf{h}_r^H)$ and the AP estimates the channel $\mathbf{h}_d$. The AP sets transmit beamforming as $\mathbf{w}^k = \sqrt{p} \frac{\mathbf{h}_d}{\|\mathbf{h}_d\|}$.
- 3: **repeat**
- 4: Let the IRS be in the receiving mode and the AP broadcast a pilot signal with transmit beamforming $\mathbf{w}^k$. The IRS estimates $\arg(\mathbf{g}_n^H \mathbf{w})$ and computes the phase shift θ^{k+1} by using (19) with given $\mathbf{w}^k$.
- 5: Let the IRS be in the reflecting mode with given θ^{k+1} and the user broadcast a pilot signal. The AP estimates the composite channel $\mathbf{h}_r^H \Theta \mathbf{G} + \mathbf{h}_d^H$ and computes the new transmit beamforming $\mathbf{w}^{k+1}$ by using (20) with given θ^{k+1} .
- 6: Update $k = k + 1$.
- 7: **until** the fractional increase of the objective value is below a threshold $\epsilon > 0$ or the maximum number of iterations is reached.

Activate Windows

Go to Settings to activate Wi

Large Intelligent Surface-Assisted Wireless Communication

Exploiting Statistical CSI - Yu

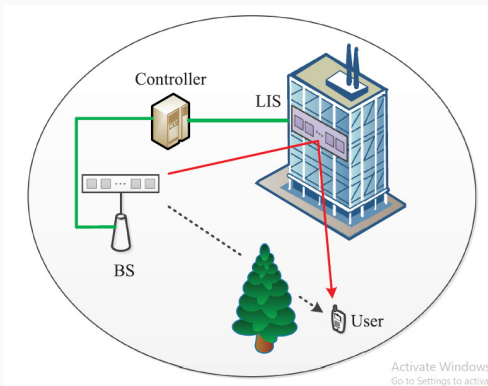
Han, Wankai Tang, Shi Jin,

Chao-Kai Wen, and Xiaoli Ma

Contributions

- Performance evaluation of an LIS assisted large scale antenna system by formulating a tight upper bound of the ergodic spectral efficiency (ESE)
- Investigated the effect of phase shifts on the ESE in different propagation scenarios
- Optimal phase shift design based on the upper bound of ESE and statistical CSI

System Model



- Rayleigh fading channel model ($\triangleq \mathbf{g}$) between the BS and UE
- Channel between BS and LIS

$$\mathbf{H}_1 = \sqrt{\frac{K_1}{K_1 + 1}} \bar{\mathbf{H}}_1 + \sqrt{\frac{1}{K_1 + 1}} \tilde{\mathbf{H}}_1,$$

- Channel between LIS and UE

$$\mathbf{h}_2 = \sqrt{\frac{K_2}{K_2 + 1}} \bar{\mathbf{h}}_2 + \sqrt{\frac{1}{K_2 + 1}} \tilde{\mathbf{h}}_2,$$

- Deterministic parts:

$$\bar{\mathbf{H}}_1 = \mathbf{a}_N^H(\theta_{\text{AoA},1}) \mathbf{a}_M(\theta_{\text{AoD},1}), \quad \bar{\mathbf{h}}_2 = \mathbf{a}_N(\theta_{\text{AoD},2})$$

- Received signal at user:

$$r = \sqrt{P}(\mathbf{h}_2 \Phi \mathbf{H}_1 + \mathbf{g}) \mathbf{f}^H s + w = \sqrt{P}(\mathbf{h} + \mathbf{g}) \mathbf{f}^H s + w$$

- MRT beamforming at the BS: $\mathbf{f} = \frac{\mathbf{h}+\mathbf{g}}{\|\mathbf{h}+\mathbf{g}\|}$
- Ergodic spectral efficiency (ESE) is given as

$$C = \mathbb{E} \left\{ \log_2 \left(1 + \frac{P}{\sigma_w^2} \|\mathbf{h}_2 \Phi \mathbf{H}_1 + \mathbf{g}\|^2 \right) \right\}.$$

- Upper bound of the ESE:

$$C \leq C^{\text{ub}} = \log_2 \left(1 + P \left(\gamma_1 \|\bar{\mathbf{h}}_2 \Phi \bar{\mathbf{H}}_1\|^2 + \gamma_2 MN + M \right) \right),$$

where

$$\gamma_1 = \frac{K_1 K_2}{(K_1 + 1)(K_2 + 1)}, \gamma_2 = \frac{K_1 + K_2 + 1}{(K_1 + 1)(K_2 + 1)}.$$

- Proof uses Jensen's inequality and computing the expectations of the individual terms inside the log term

- Effects of Rician-K factors and phase shifts studied
 - If $K_1 = 0$ or $K_2 = 0$, random phase shifts can be set
- Optimal phase shift design:

$$\Phi_{\text{opt}} = \max_{\Phi} \|\bar{\mathbf{h}}_2 \Phi \bar{\mathbf{H}}_1\|^2.$$

- Solution:

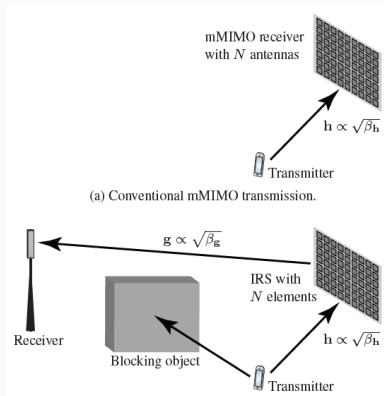
$$\phi_{\text{opt},n} = 2\pi \frac{d}{\lambda} (n-1) (\sin \theta_{\text{AoA},1} - \sin \theta_{\text{AoD},2}).$$

$$C_{\text{max}}^{\text{ub}} = \log_2 (1 + PM(\gamma_1 N^2 + \gamma_2 N + 1)).$$

- Influence of discrete phase shifts also studied (details in paper)

**Demystifying the Power Scaling
Law of Intelligent Reflecting
Surfaces and Metasurfaces - Emil
Björnson, and Luca Sanguinetti**

- It has been shown in the literature that the SNR of a single user IRS aided transmission increases as N^2 , where N is the number of passive reflecting elements
- Conventional massive MIMO provides an SNR gain of N
- This paper shows that this interpretation is incorrect and a very large number of reflecting elements is needed for an IRS to obtain SNRs comparable to massive MIMO



- Let us consider a simple channel model:

$$\mathbf{h} = \sqrt{\beta_{\mathbf{h}}} \begin{bmatrix} e^{j\phi_1} \\ \vdots \\ e^{j\phi_N} \end{bmatrix}, \quad \mathbf{g} = \sqrt{\beta_{\mathbf{g}}} \begin{bmatrix} e^{j\psi_1} \\ \vdots \\ e^{j\psi_N} \end{bmatrix}$$

- Massive MIMO: Rxed signal and SNR after MRC:

$$y_{\text{mMIMO}} = \mathbf{v}^T \mathbf{h} \sqrt{P_{\text{tx}}} s + \mathbf{v}^T \mathbf{n}, \quad \text{SNR}_{\text{mMIMO}} = \frac{\|\mathbf{h}\|^2 P_{\text{tx}}}{\sigma^2} = \frac{N \beta_{\mathbf{h}} P_{\text{tx}}}{\sigma^2}$$

- IRS: Rxed signal and SNR after choosing the phase shifts appropriately:

$$y_{\text{IRS}} = \mathbf{g}^T \Theta \mathbf{h} \sqrt{P_{\text{tx}}} s + n, \quad \text{SNR}_{\text{IRS}} = \frac{\mu^2 N^2 \beta_{\mathbf{g}} \beta_{\mathbf{h}} P_{\text{tx}}}{\sigma^2}.$$

- SNR in the IRS case should be interpreted as

$$\text{SNR}_{\text{IRS}} = \overbrace{\mu^2 N \beta_{\mathbf{g}}}^{\leq 1, \text{ Fraction of reflected power reaching receiver}} \times \underbrace{\frac{N \beta_{\mathbf{h}} P_{\text{tx}}}{\sigma^2}}_{=\text{SNR}_{\text{mMIMO}}} .$$

- Law of energy conservation needs to hold which means $\overbrace{\mu^2 N \beta_{\mathbf{g}}}^{\leq 1}$
- $N \beta_{\mathbf{h}}$ is also ≤ 1
- Therefore $\text{SNR}_{\text{IRS}} \leq \text{SNR}_{\text{mMIMO}}$

**Deep Learning for Large
Intelligent Surfaces in Millimeter
Wave and Massive MIMO
Systems - Abdelrahman Taha,
Muhammad Alrabeiah, and
Ahmed Alkhateeb**

Contributions

- A novel LIS architecture is proposed, where all the elements are passive except for a few randomly distributed active channel sensors (connected to baseband of the LIS controller)
- A DL based LIS reflection design is proposed given the knowledge of the sampled channels at the active sensors

System Model

- Single antenna Tx, Rx and LIS with M antennas, K subcarriers
- Block fading mmWave cluster channel model
- Reflection coefficients (phase shifts) chosen from a codebook
- Received signal:

$$\begin{aligned} y_k &= \mathbf{h}_{\text{R},k}^T \mathbf{\Psi} \mathbf{h}_{\text{T},k} s_k + n_k, \\ &\stackrel{(a)}{=} (\mathbf{h}_{\text{R},k} \odot \mathbf{h}_{\text{T},k})^T \boldsymbol{\psi} s_k + n_k \end{aligned}$$

- Only the channel coefficients of active sensors ($\bar{M} \ll M$) available at the LIS

Optimization Problem

$$R^* = \max_{\psi \in \mathcal{P}} \frac{1}{K} \sum_{k=1}^K \log_2 \left(1 + \text{SNR} |(\mathbf{h}_{T,k} \odot \mathbf{h}_{R,k})^T \psi|^2 \right).$$

Key Idea

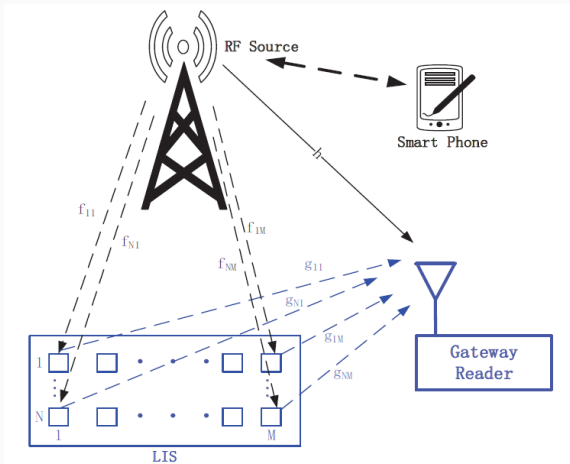
- Sampled channel estimation phase: Assume perfect channel estimates for the active channel sensors ($\bar{M} \times 1$ channel)
- Learning phase: Exhaustive search done to get the achievable rates for all the beamforming vectors in the codebook
 - Achievable rate computation uses the complete $M \times 1$ channel
 - DL Input: Sampled channel
 - DL Output: Achievable rate vector (size of codebook $\times 1$)
 - Learn the DL model
 - Prediction phase: Input the sampled channel estimate, obtain the achievable rate vector and choose the vector which gives the best rate

**Channel Estimation for Ambient
Backscatter Communications
with Large Intelligent Surface -
Hui Zheng, Ziyao Yang, Gongpu
Wang, Ruisi He, and Bo Ai**

Contributions

- Proposed an iterative channel estimator to acquire the channel parameters of an LIS assisted ambient backscatter communication system
- CRLB derived for the estimator

System Model

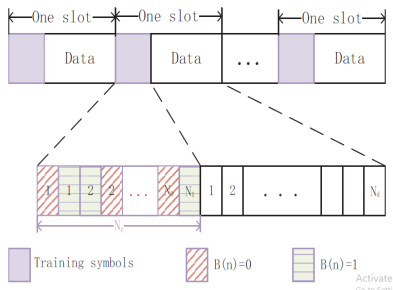


- Ambient backscatter communication with an RF source, a backscatter device (utilizes LIS to communicate), and a reader.
- $N \times M$ units in the LIS. Signals backscattered from the $(i,j)^{\text{th}}$ unit given as $r_{ij}(n) = \eta_{ij} f_{ij} g_{ij} s(n) B(n)$, where $B(n)$ is the binary information bit that the backscatter device aims to convey to the reader
- Received signal at the reader:

$$y(n) = \begin{cases} hs(n) + w(n), & B(n) = 0 \\ hs(n) + \sum_{i=1}^N \sum_{j=1}^{1M} \eta_{ij} f_{ij} g_{ij} s(n) + w(n), & B(n) = 1 \end{cases}$$

- Define $\delta = \sum_{i=1}^N \sum_{j=1}^M \eta_{ij} f_{ij} g_{ij}$ and $\mu = h + \delta$
- Goal is to estimate h and δ
- Assumption: Phases of f_{ij} and g_{ij} available at the LIS. δ will be

$$\delta_0 = \sum_{i=1}^N \sum_{j=1}^M |\eta_{ij}| \cdot |f_{ij}| \cdot |g_{ij}|.$$



- Define time indices for the training symbols when $B(n) = 0$ and $B(n) = 1$ as $\mathcal{N}_0 = \{\alpha_1, \dots, \alpha_{N_0}\}$ and $\mathcal{N}_1 = \{\beta_1, \dots, \beta_{N_1}\}$, respectively. Corresponding training sets $\mathbf{s}_0$, $\mathbf{s}_1$.
- Fixed training signal: $y_0 = h s_0 + w_0$, $y_1 = h s_1 + \delta_0 s_1 + w_1$

$$y_p = \mathbf{P} \mathbf{x} + w_p, \quad \text{where} \quad \mathbf{P} = \begin{bmatrix} s_0, & 0_{N_0 \times 1} \\ s_1, & s_1 \end{bmatrix}, \quad \mathbf{x} = [h, \delta_0]^T$$

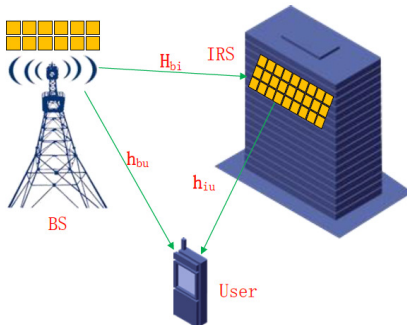
- Iterative LS estimation: Estimate h and δ_0 by LS, subtract the contribution of $\mathbf{s}_1$ from $\mathbf{y}_1$ and reestimate h and do it multiple times
- CRLB derived for the estimator

**Outage Probability Analysis and
Minimization in Intelligent
Reflecting Surface-Assisted
MISO Systems - Chang Guo,
Ying Cui, Feng Yang, and
Lianghui Ding**

Contributions

- Considered an IRS assisted downlink MISO system in an Rician fading environment where the LoS components do not change and the NLoS components vary slowly
- MRT transmission adapted to the LoS components
- Obtained an expression for the outage probability of the IRS assisted system
- Phase shifts optimized to minimize the outage probability

System Model



- LoS components of the CSI:

$$\bar{\mathbf{h}}(\Theta) \triangleq \sqrt{\frac{\alpha_{bu}\mathcal{K}_{bu}}{\mathcal{K}_{bu}+1}}\bar{\mathbf{h}}_{bu} + \sqrt{\frac{\alpha_{iu}\alpha_{bu}\mathcal{K}_{iu}}{\mathcal{K}_{iu}+1}}\bar{\mathbf{h}}_{iu}\Phi(\Theta)\bar{\mathbf{H}}_{bi}.$$

- After MRT, the channel capacity is:

$$C(\Theta) = \log_2(1 + p|g(\Theta)|^2)), \quad \text{where} \quad g(\Theta) \triangleq \mathbf{h}(\Theta) \frac{\bar{\mathbf{h}}(\Theta)^H}{\|\bar{\mathbf{h}}(\Theta)\|_2}$$

- $C(\Theta)$ is random due to the NLoS slowly varying components (not known to the BS). Outage probability:

$$P_o(\Theta) \triangleq Pr(C(\Theta) < R) = Pr\left[|g(\Theta)|^2 < \frac{2^R - 1}{p}\right]$$

- Outage probability derived and using that the phase shifts are optimized
- Optimal outage probability characterized at high SNR regime

**Reconfigurable intelligent
surfaces for energy efficiency in
wireless communication - C.
Huang, A. Zappone, G. C.
Alexandropoulos, M. Debbah,
and C. Yuen**

Contributions

- MU-MIMO downlink system where the BS signal reaches the users through a RIS with finite number of reflecting units
- Formulated the energy efficiency (EE) maximization problem to optimize the phase shifts and the DL transmit powers under max power and min QoS constraints
- Developed two low complexity algorithms to solve the EE maximization problem.
 - Frameworks of alternating maximization, sequential fractional programming and conjugate gradient search
- Numerical results show that the proposed algorithms provide higher EE performance than a traditional relay assisted (AF) communication system
- Typical resource allocation problem

More Papers

- Wireless Communications Through Reconfigurable Intelligent Surfaces
- Cramér-Rao Lower Bounds for Positioning with Large Intelligent Surfaces using Quantized Amplitude and Phase
- Performance Analysis of Large Intelligent Surfaces (LISs): Asymptotic Data Rate and Channel Hardening Effects
- Beamforming Optimization for Wireless Network Aided by Intelligent Reflecting Surface With Discrete Phase Shifts
- Intelligent Reflecting Surface Aided Multigroup Multicast MISO Communication Systems
- Utility-Based Precoding Optimization Framework for Large Intelligent Surfaces
- etc.

THANK YOU!